Research Methodology

Module 3

Sampling Techniques and Data Collection Methods

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1 Census vs. Sampling

1.1 Basic Terms

- **Population (universe):** the complete group about which we want to draw conclusions.
- **Element:** a single unit of the population (a student, a household).
- **Sampling frame:** the list from which the sample is drawn (e.g. a college roll list).
- **Sample:** a part of the population selected for study.
- **Parameter vs. statistic:** a value describing the population vs. a value calculated from the sample.

1.2 Census

A census **studies every single member** of the population. Example: counting every student of a college.

1.3 Sampling

Sampling studies **only a selected part** and uses it to draw conclusions about the whole population.

	Census	Sampling
Coverage	Entire population	A part of the population
Cost	High	Low
Time	Long	Short
Accuracy	Free of sampling error	Has sampling error, but can be small
Effort	Large workforce needed	Small team is enough
Suitable when	Population is small, or full detail is required	Population is large, or testing destroys the item
Example	National population census	Opinion poll of 1,000 voters

> **Note** A census is not always more accurate. In huge populations it can contain more **non-sampling errors** (data entry, interviewer mistakes) than a well-planned sample.

1.4 Sampling Process

1. Define the target population.
2. Prepare the sampling frame.
3. Decide the sampling method.
4. Decide the sample size.
5. Select the sample and collect data.

> **Watch Out** Two kinds of error: **sampling error** (the sample differs from the population by chance; falls as size grows) and **bias** (systematic error from poor selection; does not fall by increasing size).

2 Probability and Non-Probability Sampling

	Probability sampling	Non-probability sampling
Selection	Random; every unit has a known chance	Judgement or convenience; chance unknown
Bias	Low	Higher
Generalisation	Possible to the population	Limited
Needs frame	Yes	Not always
Cost and time	Higher	Lower

2.1 Probability Sampling Methods

Simple Random Sampling

Every member has an **equal chance** of selection, using a lottery method or random number table or software.

- **Pros:** unbiased, simple to understand.
- **Cons:** needs a complete list; may miss small subgroups.

Systematic Sampling

Choose a random start, then take every k-th unit, where

$$k = \frac{N}{n} \quad (N = \text{population size},\ n = \text{sample size})$$

Example: 500 students, sample of 50, so $k = 10$. Start at 4, then select 4, 14, 24, ...

- **Pros:** quick and easy.
- **Cons:** a hidden pattern in the list can create bias.

Stratified Random Sampling

Divide the population into **homogeneous groups (strata)** such as year or gender, then randomly sample from each.

- **Proportionate:** sample size in each stratum is proportional to its size.
- **Pros:** every subgroup is represented; more precise.
- **Cons:** needs prior information about strata.

Cluster Sampling

Divide the population into **naturally occurring groups (clusters)** such as colleges or villages, randomly choose some clusters, and study *all* (or a sample of) units inside them.

- **Pros:** cheap when the population is spread over a large area.
- **Cons:** less precise; clusters may differ from one another.

> **Exam Tip** **Stratified vs. cluster:** strata are *similar inside, different from each other* and we sample from *every* stratum. Clusters are *mini-versions of the population* and we pick *some* clusters.

Multi-Stage Sampling

Sampling is done in **stages**, from large units to small ones. Example: state → districts → colleges → students. Practical for national-level studies.

2.2 Non-Probability Sampling Methods

Convenience Sampling

Selects whoever is **easiest to reach**. Example: surveying classmates. Quick and cheap, but heavily biased.

Purposive (Judgement) Sampling

The researcher **deliberately selects** units that best serve the purpose. Example: interviewing only senior software engineers for an expert opinion.

Quota Sampling

Fixed numbers (quotas) are set for categories (e.g. 50 male, 50 female) and filled by convenience. It is like stratified sampling, but *without* random selection.

Snowball Sampling

Existing participants **refer other participants**. Useful for hard-to-reach groups such as rare-disease patients or hidden communities.

Method	Key idea
Simple random	Equal chance, pure lottery
Systematic	Every k-th unit after a random start
Stratified	Random sample from every subgroup
Cluster	Random choice of whole groups
Convenience	Whoever is easiest to get
Purposive	Chosen for a specific purpose
Quota	Fill fixed numbers by category
Snowball	Participants bring in others

> **Note** **Sample size** depends on population size, variability, required accuracy (margin of error), and confidence level. Larger samples reduce sampling error but cost more.

3 Primary and Secondary Data Collection

	Primary data	Secondary data
Meaning	Freshly collected by the researcher for the study	Already collected by someone else for another purpose
Sources	Surveys, interviews, observation, experiments	Books, journals, government records, websites, company reports
Cost and time	High	Low
Fit to study	Exactly suited	May not fit perfectly
Reliability	Known to the researcher	Must be checked
Example	Survey of BCA students	Census data, university results

> **Watch Out** Always check secondary data for **authority, date, purpose, definitions used, and accuracy** before relying on it.

4 Data Collection Methods

4.1 Survey

A survey collects information from a **sample of people** using standard questions, to describe attitudes, opinions, behaviour, or facts.

- **Modes:** face-to-face, telephone, mail, and online (e.g. web forms).
- **Pros:** reaches many people; data is easy to quantify.
- **Cons:** low response rate, shallow answers, possible dishonest replies.

4.2 Observation

The researcher **watches and records** behaviour directly, without relying on what people say.

- **Participant vs. non-participant:** the researcher joins the group, or watches from outside.
- **Structured vs. unstructured:** a fixed checklist, or free note-taking.
- **Disguised vs. undisguised:** subjects unaware or aware of being observed.
- **Pros:** actual behaviour, not reported behaviour.
- **Cons:** time-consuming, observer bias, cannot reveal opinions or reasons.

4.3 Interview

A **direct, oral exchange** between researcher and respondent, with questions asked and answers recorded.

Type	Description
Structured	Fixed questions in fixed order.
Semi-structured	Prepared guide, but the interviewer may probe further.
Unstructured	Open conversation around a topic.
Focus group	A group discussion led by a moderator.

- **Pros:** deep information, can clarify and probe, high response rate.
- **Cons:** costly, slow, interviewer bias, hard to analyse.

4.4 Questionnaire

A **written set of questions** filled in by the respondent personally, usually sent by mail, email, or online.

- **Pros:** cheap, wide reach, no interviewer bias, respondents answer at leisure.
- **Cons:** cannot be used with illiterate respondents, no probing, many non-responses.

4.5 Comparison of Methods

	Survey	Observation	Interview	Questionnaire
Data from	Many people	Behaviour	Few people, in depth	Many people
Cost	Medium	Medium	High	Low
Depth	Medium	Real actions	High	Low
Probing	Limited	No	Yes	No

> **Note** A **survey** is the overall *method*; a **questionnaire** and an **interview** are *tools* that a survey may use.

5 Designing Questionnaires and Schedules

5.1 Questionnaire vs. Schedule

	Questionnaire	Schedule
Filled by	The respondent	The enumerator, who asks and records
Delivery	Mail, email, online	Face-to-face
Literacy needed	Yes	No
Cost	Low	High
Response rate	Low	High
Clarification	Not possible	Possible

5.2 Steps in Designing a Questionnaire

1. List the objectives and the information needed.
2. Decide the target respondents and the mode of delivery.
3. Choose the question types and write the questions.
4. Arrange the questions in a logical order.
5. Write clear instructions and a short introduction.
6. Pretest with a small group, revise, and then finalise.

5.3 Types of Questions

- **Open-ended:** respondent answers freely. Rich but hard to analyse.
- **Closed-ended:** respondent picks from options (yes/no, multiple choice). Easy to analyse.
- **Rating scale:** such as the Likert scale (Strongly disagree to Strongly agree).
- **Ranking:** order items by preference.
- **Dichotomous:** only two options.
- **Contingency (filter):** asked only if an earlier answer applies.

5.4 Principles of Good Questions

- Use **simple, clear, everyday** words.
- Ask **one thing at a time**; avoid double-barrelled questions.
- Avoid **leading** or loaded wording.
- Avoid **ambiguous** terms like "often" without a definition.
- Make options **exhaustive and mutually exclusive**.
- Keep the questionnaire **short** and relevant.
- Put personal or sensitive questions **at the end**.

Fault	Poor question	Better question
Double-barrelled	Is the course interesting and easy?	Ask separately about interest and difficulty.
Leading	Don't you agree the library is excellent?	How would you rate the library?
Vague	Do you use the internet often?	How many hours per day do you use the internet?

5.5 Order and Layout

- Start with easy, interesting, non-threatening questions.
- Group related questions together, moving from general to specific.
- Place demographic questions (age, course) either at the start or end, consistently.
- Leave enough space, number the questions, and keep the layout clean.

> **Exam Tip** Always mention a **pilot test** (pretest) when answering about questionnaire design. It catches unclear questions before the main survey and is a favourite exam point.

6 Quick Revision Sheet

> - **Census:** all units. **Sampling:** part of the population; cheaper and faster.
> - **Probability sampling:** simple random, systematic, stratified, cluster, multi-stage.
> - **Non-probability sampling:** convenience, purposive, quota, snowball.
> - **Systematic:** $k = N/n$.
> - **Primary data:** fresh, costly. **Secondary data:** existing, cheaper, verify first.
> - **Methods:** survey, observation, interview, questionnaire.
> - **Questionnaire:** filled by respondent. **Schedule:** filled by enumerator.
> - **Good questions:** clear, single-idea, unbiased, ordered logically, pretested.

End of Module 3